IV. (10 points) In thermodynamic cycles, engines take heat from a hot source, output part as work and reject the rest to a cold source. Heat pumps use external work to move cold-source heat to a hot source, together with that work converted to heat. By the second law, if either working substance contacts only two reservoirs at $T_1$、$T_2$, absorbed heats $Q_1$、$Q_2$ obey $\frac{Q_1}{T_1}+\frac{Q_2}{T_2}\le 0$. Positive and negative values mean heat absorption from, and heat rejection to, a reservoir, respectively.

A heater uses a boiler at $T_0$ to heat a room directly, maintaining a constant room temperature $T_1$, above outdoors at $T_2$. Using the same energy source and the machines, what is the theoretical maximum fractional energy saving compared with direct heating?